

International Finance:

Review Session

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Review Session

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Covered Interest Arbitrage

• Disequilibrium I

- the foreign investment is more profitable than the domestic investment

$$(1 + i_t) < \frac{F_t}{S_t}(1 + i_t^*)$$

- borrow the domestic currency at i_t ($i_t \uparrow$)
- buy spot the foreign currency at S_t ($S_t \uparrow$)
- buy the foreign security at i_t^* ($i_t^* \downarrow$)
- sell forward the foreign currency F_t ($F_t \downarrow$)

- Each market will adjust until a new equilibrium is reached

Capital flows out of the domestic currency into the foreign currency

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Covered Interest Parity

- The **CIP** is a no-arbitrage condition across the spot and forward exchange rate markets, and domestic and foreign money markets

• Assumptions

- no transaction costs, no taxes and no uncertainty
- capital markets are perfect

• Equilibrium

- domestic and foreign investments with similar risk must produce identical final wealth

$$(1 + i_t) = \frac{F_t}{S_t}(1 + i_t^*)$$

- Covered interest arbitrage** is a riskless arbitrage strategy and restores any deviation from the equilibrium condition

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Covered Interest Arbitrage

• Disequilibrium II

- the domestic investment is more profitable than the foreign investment

$$(1 + i_t) > \frac{F_t}{S_t}(1 + i_t^*)$$

- rearranged as

$$\frac{S_t}{F_t}(1 + i_t) > (1 + i_t^*)$$

- borrow the foreign currency at i_t^* ($i_t^* \uparrow$)
- sell spot the foreign currency at S_t ($S_t \downarrow$)
- buy the domestic security at i_t ($i_t \downarrow$)
- buy forward the foreign currency at F_t ($F_t \uparrow$)

Capital flows out of the foreign currency into the domestic currency

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Uncovered Interest Parity

- The **UIP** is an equilibrium condition across the current and future spot exchange rate markets, and domestic and foreign money markets

- Assumptions**

- no transaction costs, no taxes and no uncertainty
- investors are risk-neutral and endowed with rational expectations

- Equilibrium**

- FX risk** arising from purchasing a foreign security is not hedged (**uncovered**)
- on average**, domestic and foreign investments must offer same **final wealth**

$$(1 + i_t) = \frac{E_t(S_{t+1})}{S_t} (1 + i_t^*)$$

- Uncovered interest arbitrage** is a speculation strategy (risky) and restores any deviation from the equilibrium condition

Uncovered Interest Arbitrage (Speculation)

- Disequilibrium I**

- on average**, foreign investment more profitable than domestic investment

$$(1 + i_t) < \frac{E_t(S_{t+1})}{S_t} (1 + i_t^*)$$

- borrow the domestic currency at i_t
- buy spot the foreign currency at S_t
- buy the foreign security at i_t^*
- leave the FX risk uncovered

If the interest rate abroad is higher than at home, future domestic currency should appreciate to guarantee a new equilibrium

Uncovered Interest Parity

- Assumptions**

- no transaction costs, no taxes and no uncertainty
- investors are risk-neutral and endowed with rational expectations

- Equilibrium**

- on average**, domestic and foreign investments offer same **final wealth**

$$(1 + i_t) = \frac{E_t(S_{t+1})}{S_t} (1 + i_t^*)$$

- gross domestic return is equal to gross uncovered foreign return

$$\frac{E_t(S_{t+1})}{S_t} = \frac{(1 + i_t)}{(1 + i_t^*)}$$

Uncovered Interest Arbitrage

- Disequilibrium II**

- on average**, domestic investment more profitable than foreign investment

$$(1 + i_t) > \frac{E_t(S_{t+1})}{S_t} (1 + i_t^*)$$

- rearranged as

$$\frac{S_t}{E_t(S_{t+1})} (1 + i_t) > (1 + i_t^*)$$

- borrow the foreign currency at i_t^*
- sell spot the foreign currency at S_t
- buy the domestic security at i_t
- leave the FX risk uncovered

If the interest rate at home is higher than abroad, future domestic currency should depreciate in the future to guarantee a new equilibrium

Law of one Price

Assumptions

- goods are homogeneous and tradeable
- no transaction costs and tariff barriers to international trade
- **commodity arbitrage** restoring driving force

Equilibrium condition

$$P_{t,i} = S_t P_{t,i}^*$$

- the price of identical goods must be equal in a common currency
- the PPP is the law of price applied to a basket of goods

Domestic and foreign currency must have the same purchasing power

Purchasing Power Parity

- The (absolute) **PPP** is an equilibrium condition

$$S_t = \frac{P_t}{P_t^*}$$

- the nominal exchange rate should be equal to price level ratio

- The relative **PPP** is a weaker equilibrium condition

$$\frac{S_{t+1} - S_t}{S_t} = \frac{\pi_{t+1} - \pi_{t+1}^*}{1 + \pi_{t+1}^*} \approx \pi_{t+1} - \pi_{t+1}^*$$

- accounts for fixed deviations caused by impediments and imperfections
- changes in the nominal exchange rate should be equal to changes in the price level ratio

Real Exchange Rate

- Real exchange rate is the price of foreign goods relative to domestic goods

$$Q_t = S_t \frac{P_t^*}{P_t}$$

- $Q_t = 1 \implies$ Competitiveness of the home country unaltered

$$S_t \frac{P_t^*}{P_t} = 1 \implies S_t P_t^* = P_t$$

- $Q_t < 1 \implies$ Competitiveness of the home country deteriorates

$$S_t \frac{P_t^*}{P_t} < 1 \implies S_t P_t^* < P_t$$

- $Q_t > 1 \implies$ Competitiveness of the home country improves

$$S_t \frac{P_t^*}{P_t} > 1 \implies S_t P_t^* > P_t$$

Real Exchange Rate

- Real exchange rate (**RER**) provides a measure of misalignment of the domestic currency versus the foreign currency

- $Q_t < 1 \implies$ home currency is **overvalued** relative to the foreign one

- $Q_t = 1 \implies$ home and foreign currency have **equal purchasing power**

- $Q_t > 1 \implies$ home currency is **undervalued** relative to the foreign one

Balance of Payments

- The balance of payments (BoP) is a **record of economic transactions** between domestic and foreign residents during a year
- **Components**
 - 1 current account (CA)
 - 2 capital account (KA)
 - 3 change in official reserves (ΔRFX)

Capital Account

- **First rule**

- the balance of payment must sum zero

$$CA + KA + \Delta RFX = 0$$

- **Second rule**

(+) $\Rightarrow$ exports of goods
inflows of domestic currency

(-) $\Rightarrow$ imports of goods
outflows of domestic currency

- **Third rule**

- the BoP is a simple accounting identity and it is mute about causation

Current Account (CA)

- The **current account** reflects the net flow of goods, services, income and unilateral transfers between domestic and foreign residents
- 1 **Balance on Goods and Services**
 - exports minus imports of goods such as commodities
 - exports minus imports of services such as shipping, banking, insurance
- 2 **Balance on Investment Income**
 - difference between receipts and payments of services of capital (interests, profits, dividends, royalties)
- 3 **Current Transfers**
 - unilateral (one side) transactions of money (foreign aid, gifts and workers' remittance)

Capital Account

- Difference between capital inflows and capital outflows

- **Capital inflows**

- decrease in foreign assets held by domestic residents
- rise in domestic assets (or foreign liabilities) held by foreigners

- **Capital outflows**

- increase in foreign assets held by domestic residents
- drop in domestic assets held by foreigners

- Capital account measures the net foreign wealth

Changes in Official Reserves

- **Changes in official reserves**

- a rise in foreign assets held by the central bank is recorded with (–)
- a fall in foreign assets held by the central bank is recorded with (+)

↑ foreign reserves $\iff$ outflow of domestic currency $\implies \Delta RFX (-)$

↓ foreign reserves $\iff$ inflow of domestic currency $\implies \Delta RFX (+)$

Current Account Deficit

- An economy running a current account deficit will experience an overall net capital inflow because foreigners on net buy domestic equities, stock, bonds and currency
- A **country is living beyond its means**
 - the overall consumption and investments exceed national savings
- A **country is an oasis of prosperity**
 - attracting investments from the rest of the world because the economy delivers higher risk-adjusted returns than other countries

International Investment Position

- The **International Investment Position (IIP)** measures the net foreign wealth of a country

$$\text{Net IIP}_t = \text{NIIP}_t = \text{Foreign Assets}_t - \text{Foreign Liabilities}_t$$

- Foreign assets $\implies$ foreign assets owned by residents
- Foreign liabilities $\implies$ domestic assets owned by foreigners
- $\text{NIIP}_t > 0 \implies$ the country is a **net creditor**
- $\text{NIIP}_t < 0 \implies$ the country is a **net debtor**

- The **NIIP** cumulates CA deficits/surpluses over time

$$\text{NIIP}_t = \text{NIIP}_{t-1} + \text{CA}_{t,t-1}$$

- $\text{CA} > 0 \implies$ home country lends internationally (foreign assets ↑)
- $\text{CA} < 0 \implies$ home country borrows internationally (foreign liabilities ↑)

Stock Approach

- The **stock** or **asset view**
 - exchange rates are financial assets used as store of value
 - what matters is the outstanding amount of currency
 - what drives the willingness of investors to hold dollar rather than yen?
- Investors are **forward looking** and the value of a currency
 - incorporates the expectations of market participants on the future course of events
 - the demand for domestic and foreign moneys depends, as the demand of any other asset, on the expected rate of return

What matters is not only what happens today but also what market expects to occur in the future

Flexible-Price Monetary Model

- **Perfect capital mobility** $\implies$ CIP holds continuously
- **Perfect substitutability** $\implies$ identical domestic and foreign assets
- **Quantity theory of money** $\implies$ money demand function
- Three main markets
 - **capital markets** $\implies$ UIP holds continuously
 - **goods markets** $\implies$ PPP holds continuously
 - **money markets** $\implies$ continuum equilibrium between demand and supply for money

Flexible-Price Monetary Model

- **Fundamental flexible-price monetary equation**

$$s_t = (m_t - m_t^*) - \kappa(y_t - y_t^*) + \lambda(i_t - i_t^*)$$

- nominal exchange rate is driven by the relative excess supply of money

- **Forward looking model**

$$s_t = \frac{1}{1+\lambda} \sum_{j=0}^{\infty} \left(\frac{\lambda}{1+\lambda} \right)^j E_t(z_{t+j})$$

- current nominal exchange rate is equal to a discounted sum of the expected value of future fundamentals

Flexible-Price Monetary Model

- **Iterate forward**
 - s_{t+1} is a function of s_{t+2} , and so on
 - at some point any further information about future z_t is redundant (steady state equilibrium)
- **Rational expectation solution**

$$s_t = \frac{1}{1+\lambda} \sum_{k=0}^{\infty} \left(\frac{\lambda}{1+\lambda} \right)^k E_t(z_{t+k})$$

The present spot exchange rate is equal to a discounted sum of the expected value of all future fundamental variables

Meese and Rogoff's Trap

- **Exchange rate forecasting**
 - economic fundamental $\implies$ **summary of the state of the economy**
 - naive random walk $\implies$ **no information about the state of the economy**
 - naive random walk better than economic fundamentals
- **Exchange rate determination puzzle**
 - how is possible that nominal exchange rate is not linked to economic fundamentals?
- **Excess volatility puzzle**
 - volatility of the exchange rate exceeds volatility of economic variables

Long-Horizon Predictability

- **Mark (1995)** considers the following long-horizon regression

$$s_{t+k} - s_t = \alpha + \beta(z_t - s_t) + \varepsilon_{t+k}$$

- where z_t

$$z_t = (m_t - m_t^*) - (y_t - y_t^*)$$

- Current exchange rate falls below its fundamental value

$$z_t - s_t > 0$$

- **foreign currency is undervalued** and should appreciate in the future

- Current exchange rate overshoots its fundamental value

$$z_t - s_t < 0$$

- **foreign currency is overvalued** and should depreciate in the future

FX Market Efficiency: Fama (1984)

- **Predictive regression** based on **FX** returns

$$s_{t+1} - s_t = \alpha + \beta(f_t - s_t) + \varepsilon_{t+1}$$

- **Predictive regression** based on **FX** excess returns

$$s_{t+1} - f_t = \alpha + \gamma(f_t - s_t) + \varepsilon_{t+1}$$

- Obtain the 2nd regression by subtracting $(f_t - s_t)$ from both sides of the 1st regression

- $\gamma = \beta - 1$

- $s_{t+1} - s_t \implies$ exchange rate return

- $f_t - s_t \implies$ forward premium

- $s_{t+1} - f_t \implies$ excess return to currency speculation

FX Market Efficiency

- **UIP** is the cornerstone condition for FX market efficiency

$$E_t(s_{t+1}) - s_t = i_t - i_t^*$$

- a positive interest rate differential should predict a depreciation of the domestic currency

- $E_t(s_{t+1}) - s_t$ is the expected future exchange rate return

- $i_t - i_t^*$ is the interest rate differential

- Since **CIP** holds, we can obtain an equivalent representation of **UIP**

$$E_t(s_{t+1}) - s_t = f_t - s_t$$

- a positive forward premium anticipates a depreciation of the domestic currency

FX Market Efficiency: Fama (1984)

- **Predictive regression** based on **FX** returns

$$s_{t+1} - s_t = \alpha + \beta(f_t - s_t) + \varepsilon_{t+1}$$

- **Predictive regression** based on **FX** excess returns

$$s_{t+1} - f_t = \alpha + \gamma(f_t - s_t) + \varepsilon_{t+1}$$

- Under the **null hypothesis** of market efficiency

- $\alpha = 0$

- $\beta = 1$ (exchange rate return is predictable, on average)

- $\gamma = 0$ (excess return to currency speculation is zero, on average)

- $\{\varepsilon_{t+1}\}$ are serially uncorrelated (you never make the same mistake!)

Forward Premium Puzzle

- Empirical evidence suggest that β is close to **minus one**
 - β is often statistically non different from zero (large standard errors)
 - UIP-based predictive regressions fails to outperform a RW in OOS
 - as long as β is different from unity, **forward premium puzzle** remains
- What is the **economic implication** if you set $\beta = 0$?

$$E_t(s_{t+1} - s_t) = 0$$

- exchange rate return is unpredictable and is not offset by the interest rate differential
- The basis of the **carry trade strategy**
 - exploiting positive interest rate differentials across countries
 - borrowing funds in a low-interest rate currency (**funding currency**)
 - investing in a high-interest rate currency (**target currency**)

Carry Trade Strategy

- **Case 1:** as $i_t^* > i_t$ you borrow funds at home and lend abroad
 - at time t : **expected excess return**

$$\begin{aligned} E_t(ER_{t+1}) &= i_t^* - i_t + E_t(s_{t+1} - s_t) \\ &= i_t^* - i_t \end{aligned}$$

- at time $t + 1$: **realized excess return**

$$ER_{t+1} = i_t^* - i_t + (s_{t+1} - s_t)$$

- The strategy turns out to be profitable, on average, if

$$i_t^* - i_t > (s_t - s_{t+1})$$

the interest rate differential is larger than the foreign currency depreciation (investment currency)

Forward Premium Puzzle

- If $\beta = 1$ (statistically)
 - uncovered interest parity holds
 - exchange rate return is predictable, on average
 - excess return to currency speculation is equal zero, on average
- If $\beta = 0$ (statistically)
 - uncovered interest parity does not hold
 - exchange rate return is unpredictable, on average
 - excess return to currency speculation is different from zero, on average

Carry Trade Strategy

- **Case 2:** as $i_t > i_t^*$ you borrow funds abroad and lend at home
 - at time t : **expected excess return**

$$\begin{aligned} E_t(ER_{t+1}) &= i_t - i_t^* - E_t(s_{t+1} - s_t) \\ &= i_t - i_t^* \end{aligned}$$

- at time $t + 1$: **realized excess return**

$$ER_{t+1} = i_t - i_t^* - (s_{t+1} - s_t)$$

- The strategy turns out to be profitable, on average,

$$i_t - i_t^* > (s_{t+1} - s_t)$$

the interest rate differential is larger the foreign currency appreciation (funding currency)

Crash Risk

- Brunnermeier, Nagel and Pedersen (2009) suggest that
 - carry traders are subject to crash risk
 - exchange rate movements between high-interest-rate and low-interest-rate currencies are negatively skewed
 - this negative skewness is due to sudden unwinding of carry trades
 - this occurs in periods in which risk aversion increases and funding liquidity decrease
- Carry trade returns (excess returns to currency speculation)
 - **compensation for bearing crash risk**
 - a small probability that a negative event will happen and wipe out a positive excess return
 - *go up by the stairs but occasionally come down by the elevator*
 - *picking up pennies in front of a truck*

Crash Risk

- Unwinding their carry trade positions is due to losses, capital redemptions, increased margin, or reduced risk tolerance. It is associated with an increase in VIX index and TED spread
- **VIX Index**
 - measures the implied volatility of S&P 500 index
 - measures the market's expectation of stock market volatility over the next 30 day period
 - the VIX index tend to increase when risk aversion increases
 - **a proxy for global risk aversion or risk appetite**

Crash Risk

- **TED Spread**
 - difference between 3m US dollar LIBOR rate and 3m T-Bill rate
 - **LIBOR** reflects uncollateralized lending in the interbank market (subject to default risk)
 - **T-Bill** rate is riskless since it is guaranteed by the U.S. government
 - when banks face liquidity problems the TED spread typically increases
 - **a proxy for funding liquidity risk**